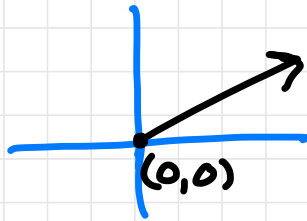


Vecteurs:

Physics



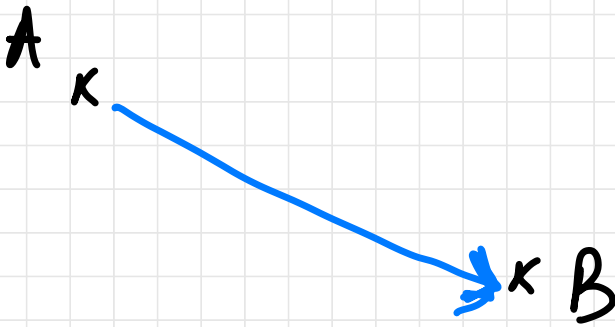
Math



CS

$\begin{pmatrix} \text{name} \\ \text{age} \end{pmatrix}$

$$\vec{AB} = \vec{CD} = \vec{u}(d, 0, w)$$



$\vec{AB}$

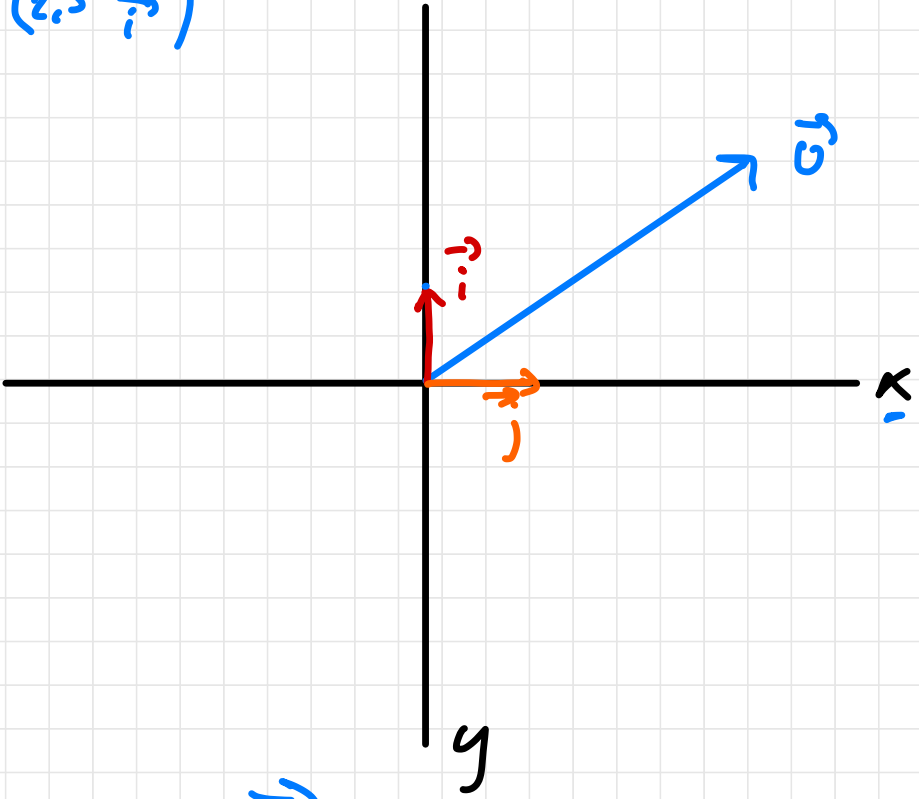
- direction
- orientation
- length, norm

C_x

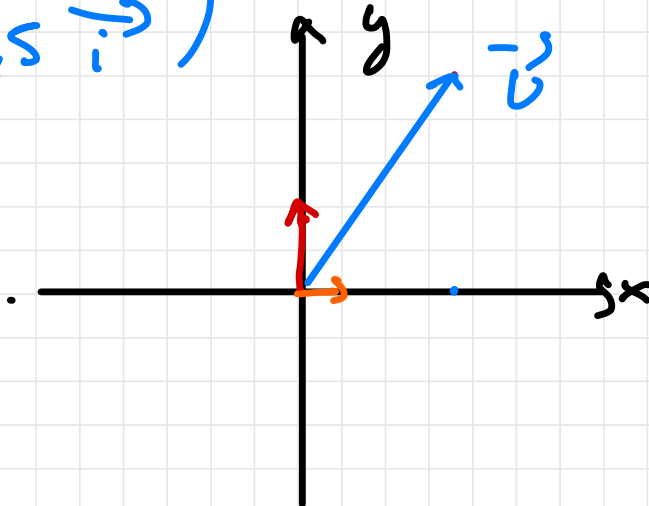
$\vec{CD}$



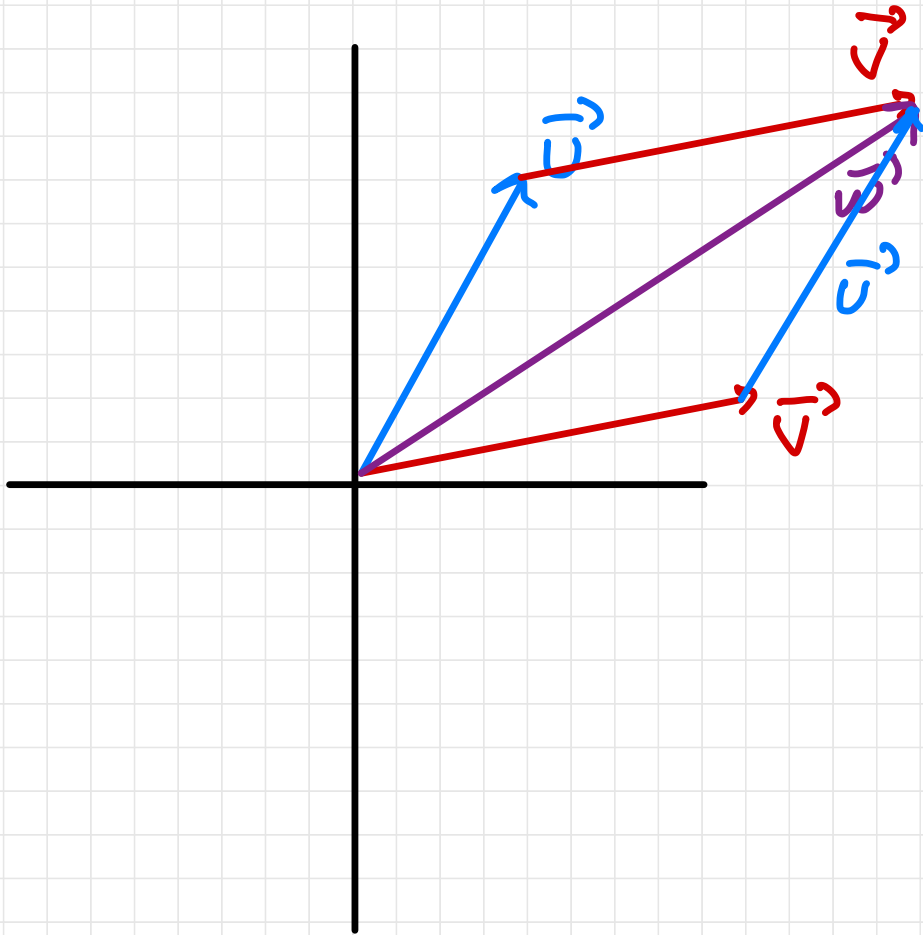
$$\vec{v} = \begin{pmatrix} 3,5 & \vec{j} \\ 2,5 & \vec{i} \end{pmatrix}$$



$$\vec{v} = \begin{pmatrix} 3,5 & \vec{j} \\ 2,5 & \vec{i} \end{pmatrix}$$



Addition :

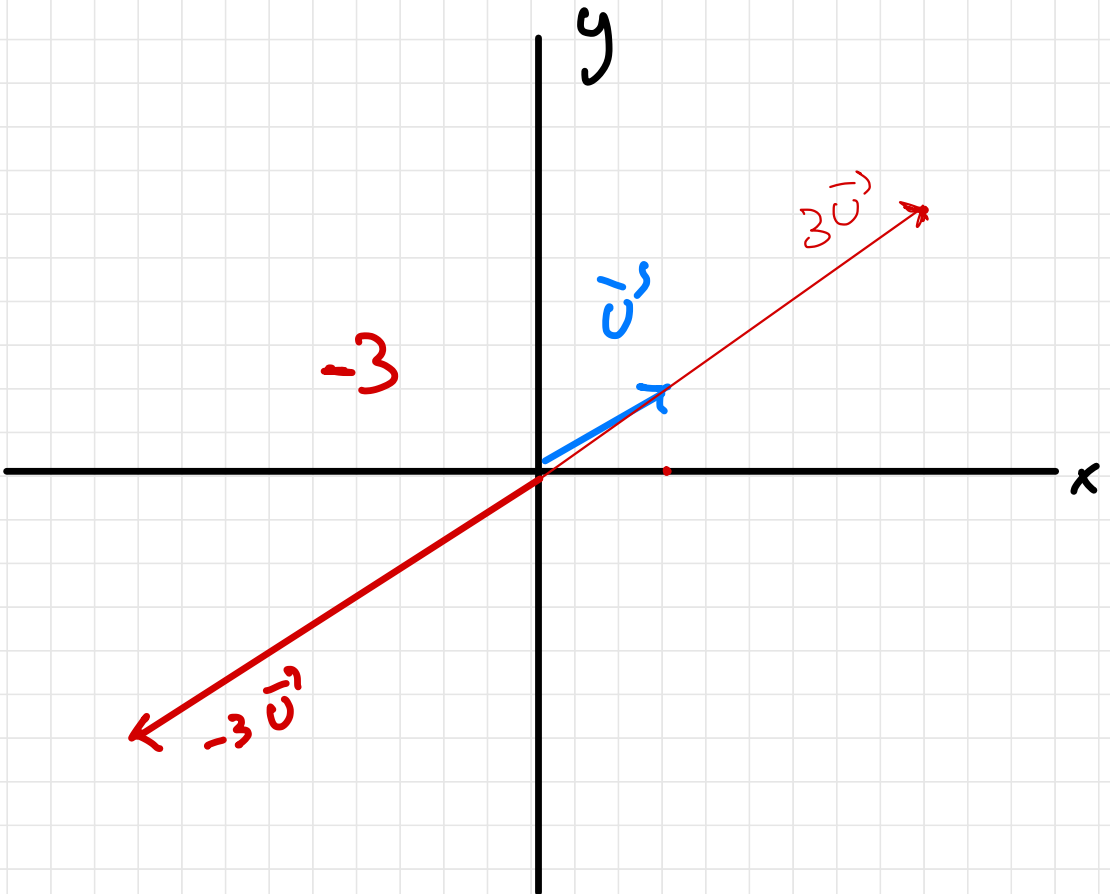


$$\vec{U} = \begin{pmatrix} u_1^4 \\ u_2^7 \end{pmatrix} \quad \vec{V} = \begin{pmatrix} v_1^9 \\ v_2^2 \end{pmatrix}$$

$$\vec{W} = \vec{U} + \vec{V} = \begin{pmatrix} u_1^4 + v_1^9 \\ u_2^7 + v_2^2 \end{pmatrix} \begin{matrix} 13 \\ 9 \end{matrix}$$

Scalar multiplication:

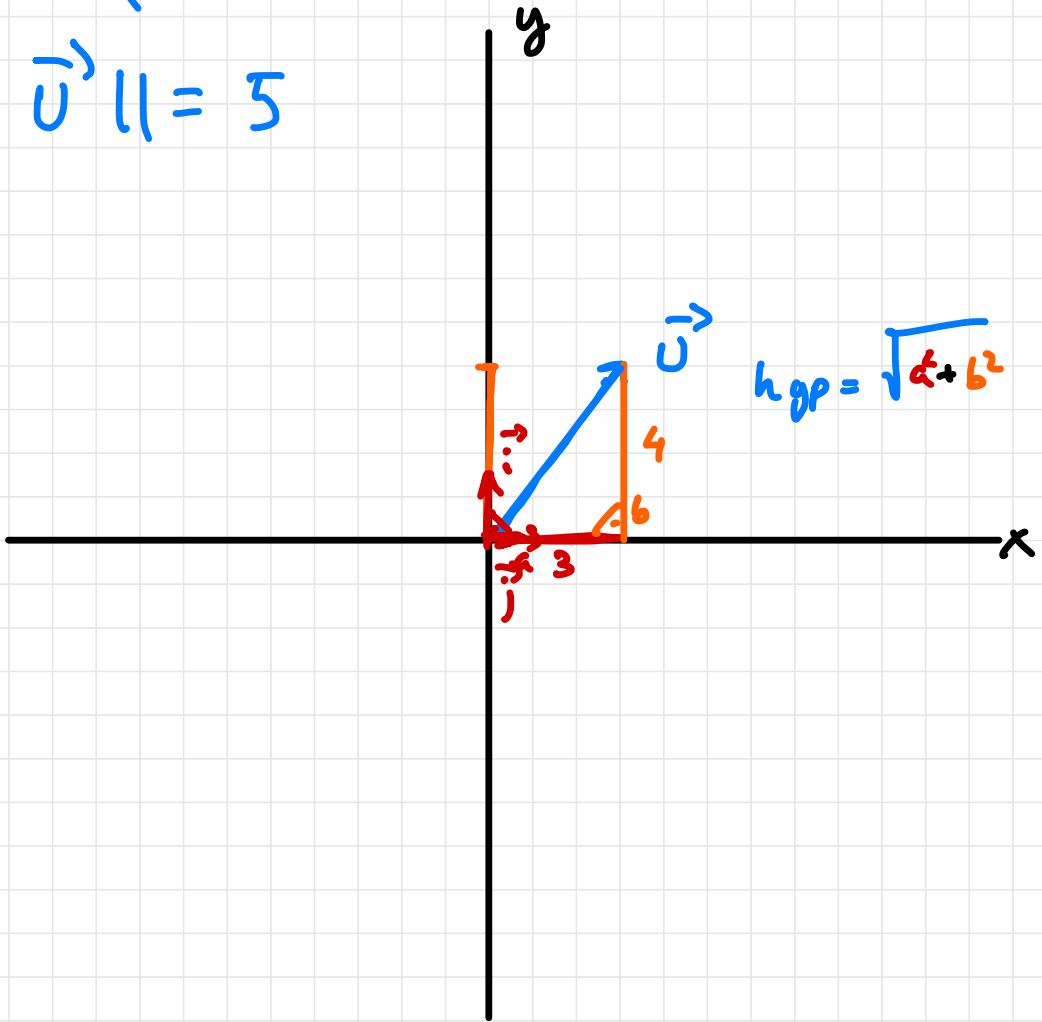
$$s \cdot \vec{u} = \begin{pmatrix} s \cdot u_1 \\ s \cdot u_2 \end{pmatrix}$$



norm:

$$\vec{u} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$\|\vec{u}\| = 5$$



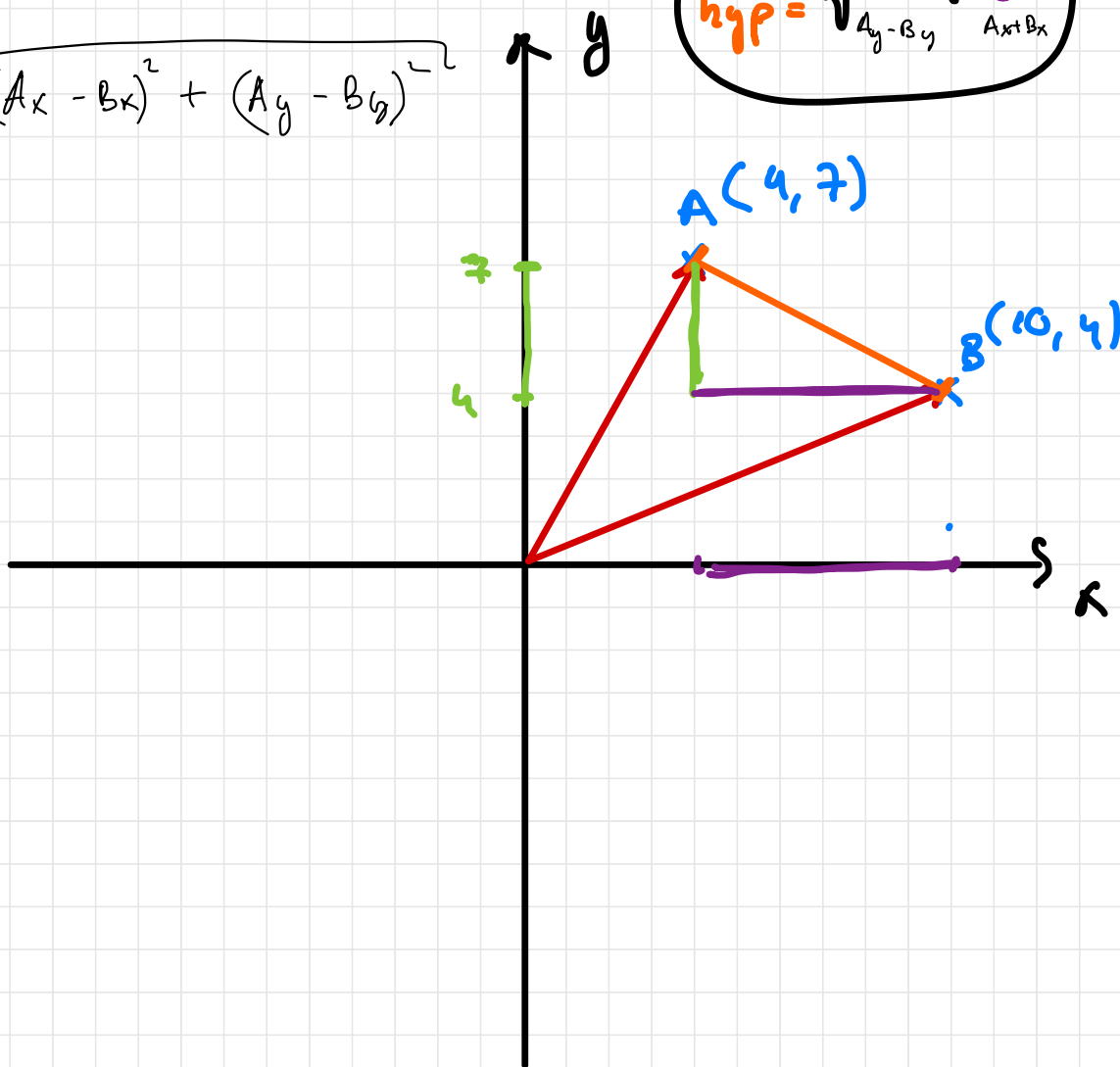
Distances:

euclidian distance

L2 - norm

$$\sqrt{(A_x - B_x)^2 + (A_y - B_y)^2}$$

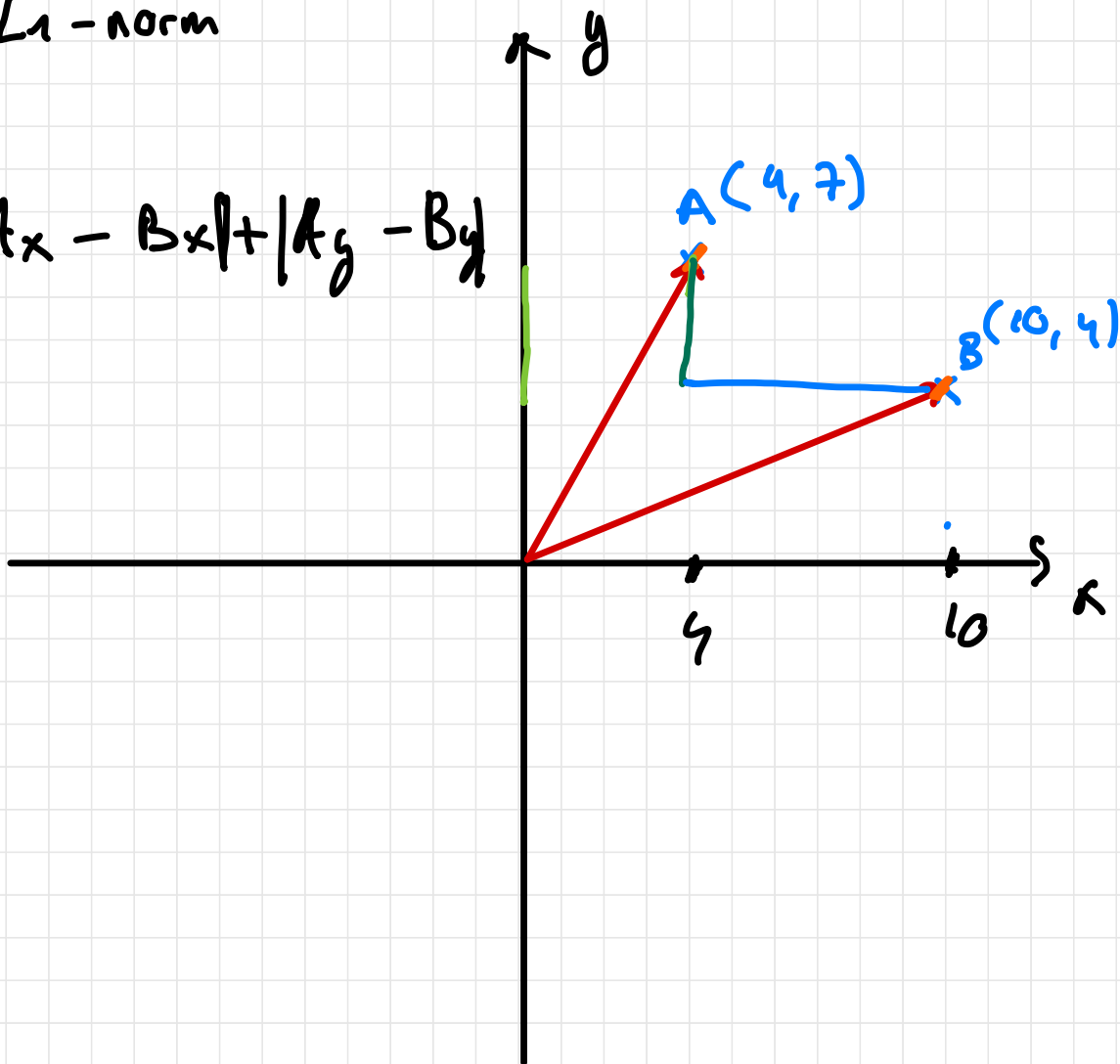
$$\text{hyp} = \sqrt{A_y - B_y + A_x - B_x}$$

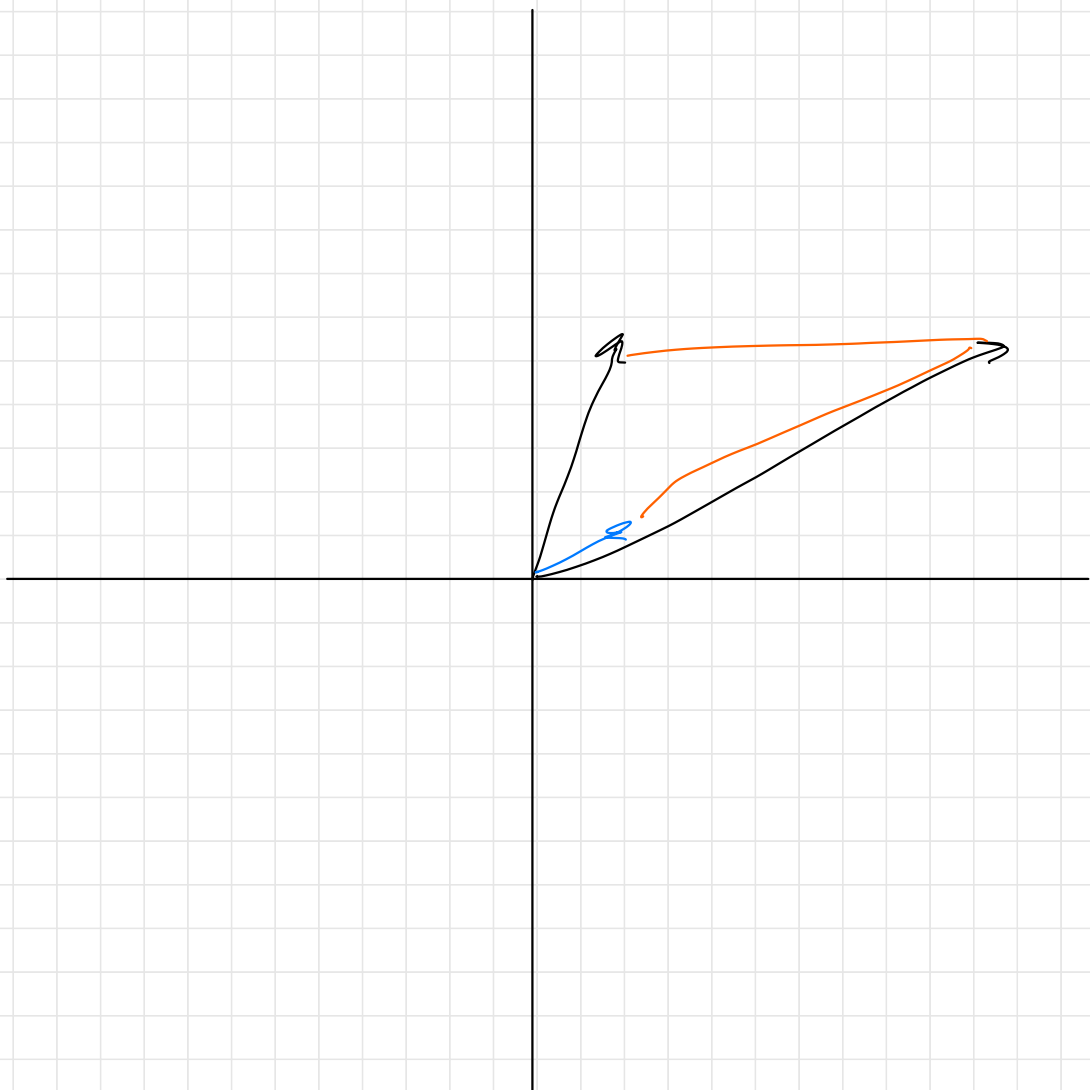


Manhattan distance

L_1 -norm

$$|A_x - B_x| + |A_y - B_y|$$





$$\vec{U} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

$$\vec{U} \cdot \vec{V} = (u_1 \cdot v_1) + (u_2 \cdot v_2)$$

$$\vec{V} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

if $\vec{U}$ and $\vec{V}$ are
orthogonal (perpendicular)
the dot product is 0

